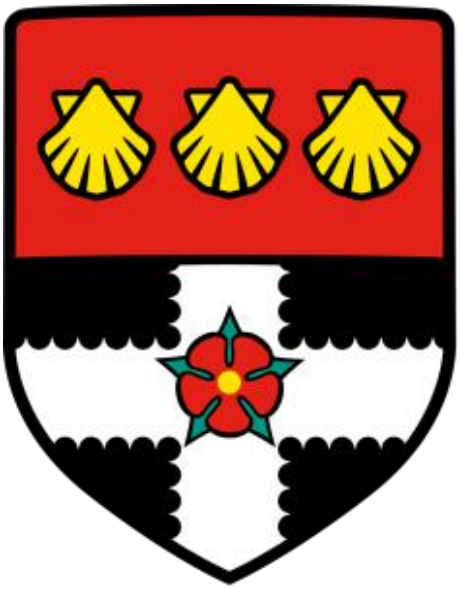
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SCHOOL OF BIOLOGICAL SCIENCES

EFFECTS OF VITAMIN B12 SUPPLEMENTATION ON SENSORY-EVOKED LOCAL FIELD POTENTIALS IN THE RAT BARREL CORTEX

UCHE DUMZO-AJUFO

BIOMEDICAL ENGINEERING

YING ZHENG

Dissertation Declaration

BI3RP3 Research Project - 2025

1. **I confirm that any contribution of results from others that are presented in this report have been acknowledged and that otherwise this report is entirely my own work.**

2. **My main inputs into my project were:**

Research question:

- ☐ I formulated my own research question, with help and guidance from my supervisor.
- ☑ I addressed a predetermined research question.

Data (tick multiple boxes if your data came from multiple sources):

- ☐ I generated my own data in the lab or the field or collected my own data from the published literature.
- ☑ I used a pre-existing dataset either from the literature, provided by my supervisor, or generated by other lab members.

Analysis:

- ☐ I designed and carried out my own analyses, with help and guidance from my supervisor.
- ☑ I carried out analyses designed by my supervisor.

3. **GenAI tools statement:**

I acknowledge the use of generative AI tools in the development of this dissertation. Specifically, ChatGPT-4o (OpenAI, https://chatgpt.com/) was used to assist with document structuring, coding tasks, and checks for spelling, punctuation, and grammar. All research, data analysis, interpretation, and written content are entirely my own. I take full responsibility for the accuracy and integrity of the work presented.

Date: 04/05/2025

Signature: 

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ABSTRACT

Vitamin B12 is an essential micronutrient involved in neural function, particularly in myelin synthesis, DNA repair, and neurotransmitter regulation (Hosseini et al., 2024). While B12 deficiency is associated with cognitive and sensory impairments, its influence on sensory cortical processing in healthy brains remains poorly understood. This study investigated the effects of B12 supplementation on neural activity in the rat barrel cortex using whisker pad stimulation and local field potential (LFP) analysis.

LFP responses were recorded in vivo from control and B12-supplemented rats (n = 7 per group) across four stimulation intensities (1.2 mA, 0.8 mA, 0.6 mA, and 0.4 mA). Key LFP features including peak amplitude, peak to peak (P2P) amplitude, and latency were analysed using ANOVA and post hoc tests. B12 supplementation significantly increased response amplitudes and reduced latencies at lower stimulus intensities, suggesting enhanced cortical excitability and more efficient sensory processing under weaker stimulation.

These findings highlight an intensity-dependent effect of B12 supplementation on sensory-evoked cortical responses. This suggests potential therapeutic applications for B12 in modulating sensory function, with broader implications for neurological health, particularly in conditions involving sensory deficits. Future studies should delve into the underlying mechanisms and assess whether similar effects occur in other sensory systems or in awake, behaving models.

TABLE OF ABBREVIATIONS

Abbreviation	Definition
ANOVA	Analysis of Variance
B12	Vitamin B12
d	Cohen's *d* (effect size measures)
Hz	Hertz (frequency unit)
LFP	Local Field Potential
mA	Milliampere (current unit)
ms	Millisecond (time unit)
mV	Millivolt (voltage unit)
P2P	Peak-to-Peak (Amplitude)
p	Probability Value (statistical significance)
s	Second (time unit)
SEM	Standard Error of the Mean
V	Volt (voltage unit)

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1 INTRODUCTION

1.1 BACKGROUND

Understanding how the brain processes sensory information is fundamental to neuroscience. Sensory processing, defined as the brain's ability to interpret environmental stimuli, is crucial for organisms to interact with and respond to their surroundings. A central structure involved in tactile sensory processing is the somatosensory cortex, particularly the barrel cortex in rodents.

The barrel cortex is organised into discrete anatomical structures known as "barrels," each corresponding to a specific whisker on the rodent's face, thereby providing a precise somatotopic map of tactile input (Petersen, 2007). This arrangement allows accurate encoding and interpretation of sensory information, making the barrel cortex a widely used model for studying cortical processing and sensory integration (Perronnet et al., 2016). Whiskers are critical for rodents' ability to navigate their environment, detect objects, and communicate (Arakawa and Erzurumlu, 2015). Deflections of a whisker are rapidly transmitted to the corresponding barrel, where complex neural circuits process these signals and generate behavioural responses.

The functionality of these neural circuits is influenced by a range of physiological factors, including micronutrient availability. Vitamin B12, an essential micronutrient, plays a crucial role in neural health through its involvement in myelin synthesis, neurotransmitter regulation, and DNA synthesis (Miller et al., 2005). Myelin, the protective sheath surrounding neuronal axons, is necessary for the rapid conduction of electrical impulses. Deficiencies in B12 can result in demyelination, impairing the speed and efficiency of neural communication, and potentially leading to significant neurological dysfunction, including impaired sensory processing (Jatoi et al., 2020).

Although the importance of B12 in maintaining general neural health is well documented, its specific role in modulating sensory-evoked cortical responses, particularly within primary sensory regions like the barrel cortex, remains relatively unexplored. Considering the established relationship between myelin integrity, synaptic plasticity, and sensory processing, it is possible that B12 supplementation could enhance neural response efficiency at the cortical level.

1.2 LITERATURE REVIEW

1.2.1 B12 and Neural Function

B12 plays a vital role in the maintenance of the central nervous system, primarily through its involvement in the formation of myelin sheaths. Myelin is critical for the rapid transmission of electrical impulses between neurons, and its integrity is essential for efficient sensory and cognitive processing. Deficiencies in B12 compromise myelin integrity, leading to slower neural conduction, impaired cognition, and, in severe cases, irreversible neurological damage.

Beyond myelin production, B12 supports the methylation of neurotransmitters such as dopamine and serotonin (Tardy et al., 2020), which are fundamental for regulating mood, cognition, and sensory perception (Rogers, 2010). B12 also facilitates the synthesis of DNA and RNA (Halczuk et al., 2023), which are processes necessary for neuronal repair and regeneration (Bader et al., 2020). These multiple roles underline the potential for B12 to influence sensory processing pathways in the brain, although the direct effects of supplementation on sensory circuits have not been thoroughly investigated (Leishear et al., 2012).

1.2.2 Food Sources of B12

B12 is predominantly obtained from animal-derived foods such as meat, fish, dairy products, and eggs (Obeid et al., 2019). Organ meats, such as beef liver and clams, are particularly rich sources (Vogeler et al., 2022). Although fish, poultry, and dairy also provide significant amounts, individuals following vegetarian or vegan diets are at higher risk of deficiency due to the absence of naturally occurring B12 in plant-based foods.

For individuals adhering to plant-based diets, fortified foods such as cereals, nutritional yeast, and plant-based milk alternatives offer a necessary source of B12 (Niklewicz et al., 2022). However, the levels found in fermented plant foods like tempeh and miso are typically insufficient to meet recommended daily intakes (Neufingerl and Eilander, 2021). Given that the daily requirement for adults is approximately 2.4 micrograms (Bor et al., 2010), supplementation is often necessary, particularly in populations with impaired absorption or restricted dietary intake, such as the elderly (Allen, 2008).

1.2.3 Sensory Processing in the Barrel Cortex

The barrel cortex serves as a suitable model for studying sensory processing due to its distinct anatomical organisation and direct communication with whisker-based sensory input. Each barrel within the cortex processes input from an individual whisker, enabling precise mapping of tactile stimuli onto cortical regions. This unique organisation enables detailed studies of how sensory information is encoded and processed within the brain (Petersen, 2007).

Neural circuits within the barrel cortex consist of networks of excitatory and inhibitory neurons, which interact to regulate responses to sensory stimuli. Local field potentials (LFPs), representing the summed electrical activity of neural populations, are widely used to investigate sensory processing within this system. The analysis of LFPs provides critical information about the timing and strength of cortical responses, with parameters such as peak amplitude and latency offering valuable insights into the efficiency of neural signalling (Greene et al., 2023).

Previous electrophysiological studies have identified stereotyped patterns of activity following whisker stimulation, including identifiable components such as the P1 and N1 peaks (Bruyns-Haylett et al., 2017). These early components reflect initial cortical processing of sensory information and provide a reliable framework for assessing the effects of experimental interventions on sensory responsiveness (Kajikawa and Schroeder, 2011).

1.2.4 B12 and Sensory Processing in the Barrel Cortex

Although the neurological effects of B12 on cognition and general brain health are well understood, little research has focused specifically on its influence within primary sensory cortices. Given its roles in promoting myelin integrity and supporting neurotransmitter regulation, B12 supplementation may enhance neural processing efficiency and plasticity within sensory systems.

Moreover, B12 has been implicated in modulating long-term potentiation (LTP), a fundamental process underlying synaptic strengthening, learning and memory (Rathod et al., 2016). Enhancing LTP could lead to improved sensory encoding and processing, suggesting that B12 may influence not only cognitive functions but also basic sensory perceptual mechanisms. However, the specific impact of B12 supplementation on sensory-evoked responses in the barrel cortex remains an area requiring further study.

1.3 RATIONALE FOR THE STUDY

Given the roles of B12 in supporting myelination, synaptic plasticity and neurotransmitter regulation, it is reasonable to hypothesise that supplementation could improve sensory processing efficiency in the brain. However, despite strong theoretical support, there is limited empirical evidence addressing this question directly, particularly within sensory regions such as the barrel cortex.

The barrel cortex's clear somatotopic mapping of sensory input, combined with the ability to measure neural activity using LFPs, presents an ideal model system to investigate the effects of B12 on sensory processing. Furthermore, rats serve as an established model in sensory neuroscience research due to the similarity of their somatosensory systems to those of other mammals, enhancing the translational relevance of findings from rodent models to broader biological contexts (Ellenbroek and Youn, 2016).

1.4 AIMS AND HYPOTHESIS

The primary aim of this study is to investigate the effects of B12 supplementation on neural activity in the barrel cortex of rats in response to whisker pad stimulation. By measuring sensory-evoked LFPs, it seeks to determine whether B12 supplementation modulates cortical responses.

It is hypothesised that rats supplemented with vitamin B12 will exhibit enhanced sensory-evoked neural responses, as reflected by increased peak amplitudes and reduced latencies in local field potentials, compared to control rats. These findings are expected to offer novel insights into the role of micronutrient status in influencing sensory processing within cortical neural circuits.

2 MATERIALS AND METHODS

2.1 ANIMALS AND ETHICAL APPROVAL

Nineteen adult female neurotypical Lister Hooded rats (aged 2-4 months, weighing 190-260 g) were used in this study. The choice of strain, sex and age was based on their suitability for cortical electrophysiological recordings, as supported by previous research (Kang et al., 2019). To minimise hormonal variability and promote consistency across electrophysiological recordings, female rats were selected for this study (Smarr and Kriegsfeld, 2022). Ethical approval for the study was granted by the University of Reading's Research Ethics Committee, and all procedures complied with the UK Home Office Animals (Scientific Procedures) Act, 1986.

The rats were housed in a temperature-controlled environment maintained at 21-23 °C with a 12-hour light/dark cycle. Standard environmental enrichment (e.g., nesting materials, shelters) was provided to promote animal welfare. All animals underwent a three-day acclimatisation period prior to the start of experimental procedures.

2.2 EXPERIMENT GROUPS AND DIET

Rats were randomly assigned to one of two experimental groups: the Control group (n = 9), which received standard tap water, and the B12 group (n = 10), which received tap water supplemented with cyanocobalamin (Sigma-Aldrich). The cyanocobalamin dosage was 50 µg/kg body weight (dry food equivalent), approximately 80 times the standard daily requirement for rats (Greibe et al., 2018). Supplementation was administered orally over a five-month period. All rats had ad libitum access to standard commercial rodent chow throughout the study.

Routine monitoring ensured that no adverse effects, such as weight loss or signs of toxicity, were observed in the rats during the study period. No deviations from expected health outcomes were recorded.

2.3 Surgical Procedure and Stimulation Protocol

At the end of the supplementation period, rats were anaesthetised with an intraperitoneal injection of urethane (1.25 g/kg), a long-lasting anaesthetic suitable for electrophysiological recordings (Shumkova et al., 2021). A 16-channel multilaminar microelectrode array was implanted perpendicularly into the barrel cortex, identified using standard neuroanatomical landmarks.

LFPs were recorded across multiple cortical layers. Sensory stimulation was applied via electrical pulses delivered to the right whisker pad at four current intensities: 1.2, 0.8, 0.6 and 0.4 mA. Each trial consisted of a train of 10 pulses delivered at 5 Hz. Although each stimulus train lasted 15 seconds, only the first 2 seconds were used for analysis to capture early LFP responses. Each rat underwent 50 trials per stimulation intensity, resulting in 200 trials per animal. Data were recorded at a sampling rate of 24,414 Hz to ensure high temporal resolution.

2.4 Data Processing

Raw LFP data were processed using MATLAB R2024b (MathWorks, USA). To enhance the signal-to-noise ratio, LFP signals from the 50 trials at each stimulus intensity were averaged for each rat. Rats with first pulse amplitudes below 0.01 mV at the lowest stimulation intensity (0.4 mA) were excluded from further analysis due to insufficient LFP responses for reliable measurement.

The following LFP parameters were extracted: peak amplitude (maximum voltage change, indicative of neural activation strength), peak-to-peak (P2P) amplitude (total response magnitude) and peak latency (time interval from stimulus onset to peak response, reflecting neural processing speed). These parameters were measured for each rat at all stimulation intensities (Brown, McAnally and Clark, 1997; Warzecha and Egelhaaf, 2000).

These parameters were analysed across three time phases: the first pulse (0-0.2 s), representing the initial cortical response; the second pulse (0.2-0.4 s), reflecting early adaptation; and the steady-state response (1.4-2.0 s), calculated as the mean response to the 8th, 9th and 10th pulses, which assessed cortical activity in response to sustained input (Brenner et al., 2009).

2.5 STATISTICAL ANALYSIS

Statistical analyses were conducted using Microsoft Excel and MATLAB. A parametric two-way repeated measures ANOVA (Excel 'Anova: Two-Factor with Replication' tool) was used to assess the main effects of group (Control vs. B12) and stimulation intensity (1.2, 0.8, 0.6 and 0.4 mA), as well as their interaction. This analysis assumed normally distributed data.

The assumption of normality was tested using the Jarque-Bera test (MATLAB jbtest function). Post-hoc pairwise comparisons at each stimulation intensity were performed using two-sample t-tests (Excel T.TEST function) on parameters that showed significant effects in the two-way ANOVA, to identify which intensities contributed to the observed significance. For significant post-hoc results, effect sizes were calculated using Cohen's *d* and interpreted as small (0.2), medium (0.5) or large (0.8) effects (Sullivan and Feinn, 2012). A significance threshold of $\alpha = 0.05$ was applied to all statistical tests.

3 RESULTS

3.1 DATA QUALITY AND SUBJECT EXCLUSIONS

Following data preprocessing, 14 rats (n = 7 per group) were included in the final analysis. Five rats were excluded due to minimal or absent responses at the 0.4 mA stimulus intensity: Rats 2 and 9 from the control group, and Rats 4, 7, and 9 from the B12 group.

Not all parameters met the assumption of normality, as shown in Table 1; however, parametric statistical analyses were still applied. This decision was justified by the robustness of parametric tests to moderate deviations from normality, the absence of extreme outliers, and the balanced experimental design following exclusions.

Table 1. Jarque-Bera test results for normality. *P-values are shown for each parameter dataset, rounded to four significant figures. Parameters with p-values ≤ 0.05, indicating a significant deviation from normality, are marked with an asterisk (*).*

Phase	Peak		P2P		Latency	
	Control	B12	Control	B12	Control	B12
First Pulse	0.1496	0.1586	0.1621	0.1572	0.3807	0.0105*
Second Pulse	0.0059*	0.2742	0.0117*	0.2907	0.001*	0.5
Steady-State	0.001*	0.0013*	0.001*	0.0359*	0.001*	0.0062*

3.2 REPRESENTATIVE LFP RESPONSES

A schematic of the stimulation protocol is shown in Figure 1, illustrating the timing of the first pulse, second pulse, and steady-state response phases within a stimulus train. The first pulse is highlighted in red, the second in blue, and the steady-state in green. This visualisation aids understanding of the temporal structure of the analysed LFP components.

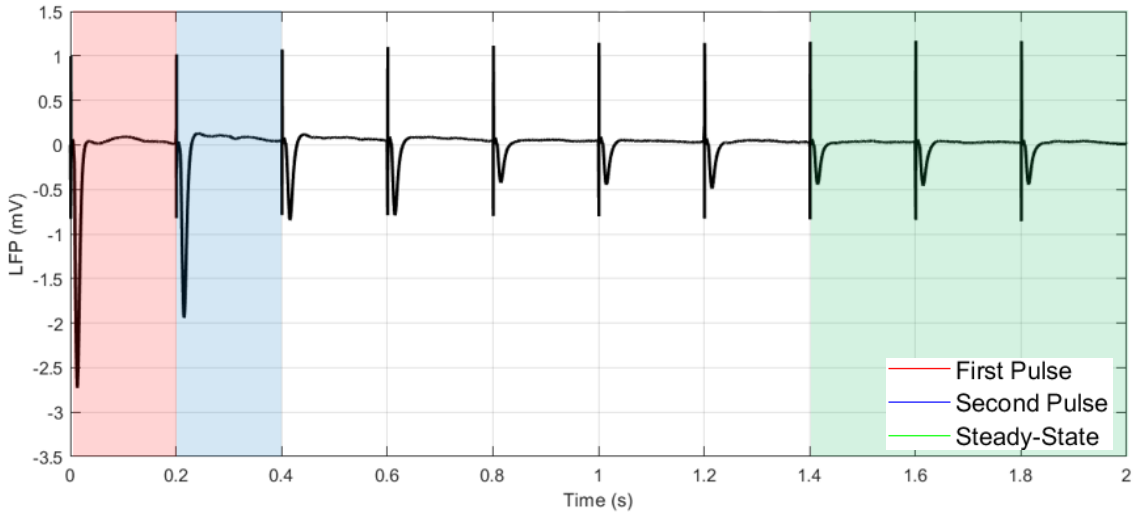


Figure 1. *Schematic of the stimulation protocol.* *The plot shows average LFP responses across 50 trials from a representative rat (Rat 1, B12 group) at 1.2 mA. The first pulse (0-0.2 s) is in red, the second (0.2-0.4 s) in blue, and the steady-state phase (1.4-2.0 s) in green. Time (s) is on the x-axis, LFP response (mV) on the y-axis.*

To illustrate overall LFP response trends within each group, group-averaged LFP waveforms at the 1.2 mA stimulus intensity (prior to exclusions) are shown in Figure 2. Both groups exhibited periodic deflections corresponding to stimulus pulses, characterised by a sharp negative peak followed by a slower recovery. The waveform shape, amplitude, and timing were broadly comparable between groups, with consistent responses maintained throughout the 2-second trial window.

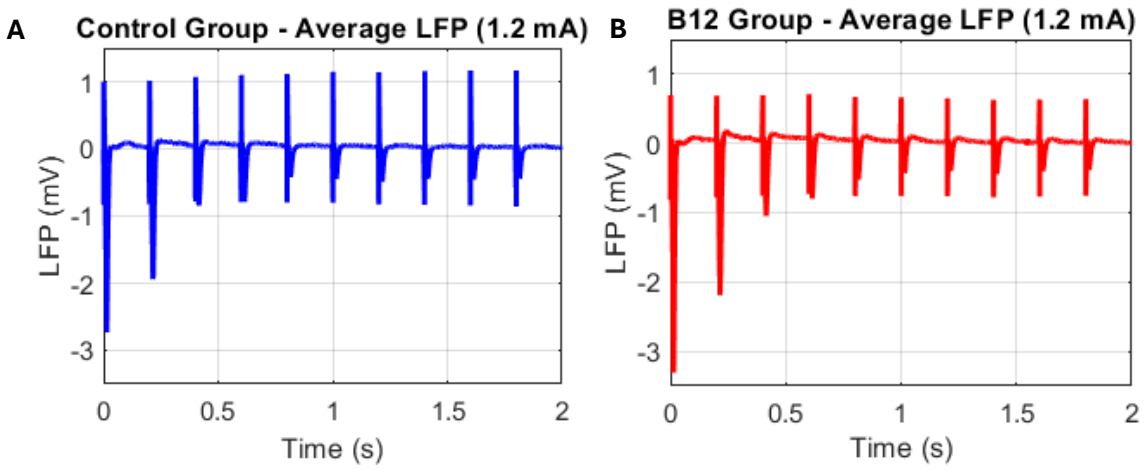


Figure 2. Group-averaged LFP responses at 1.2 mA stimulation. *Panel A: control (blue); Panel B: B12 (red), including data from rats later excluded from analysis. Time (s) is shown on the x-axis, and LFP response (mV) on the y-axis.*

3.3 Inter-Subject Variability In LFP Responses

Average responses from individual rats at each stimulation intensity are shown in Figures 3-5. These plots highlight substantial variability in response amplitude and timing across animals and intensities, particularly within the three defined response phases. This variability emphasises the importance of conducting group-level statistical analyses to identify consistent patterns and treatment-related effects.

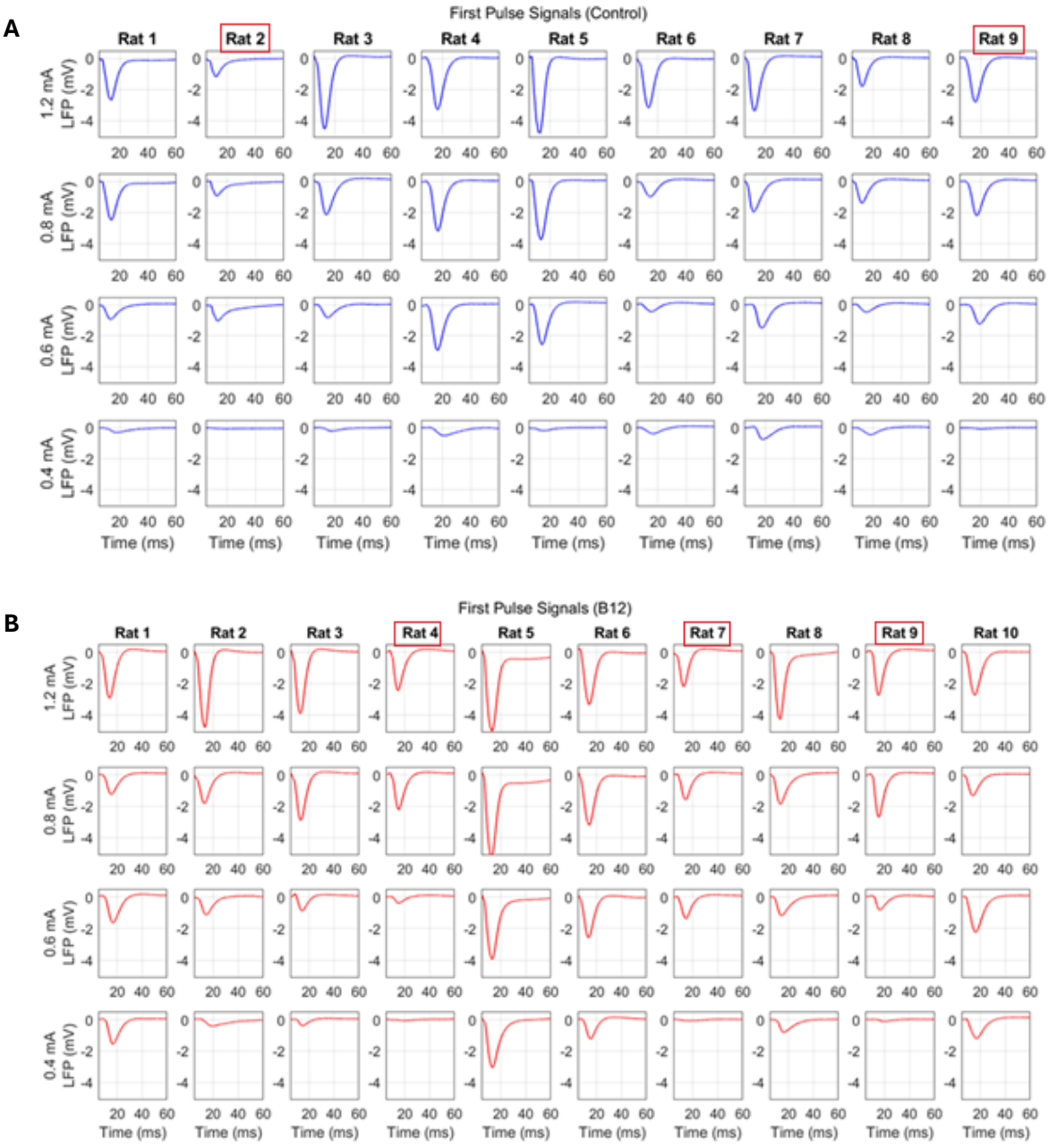


Figure 3. Individual LFP responses during the first pulse phase. *Panel A: control rats (blue); Panel B: B12-supplemented rats (red), across all intensities. Red-boxed titles mark excluded rats. Time (ms) on x-axis, LFP (mV) on y-axis.*

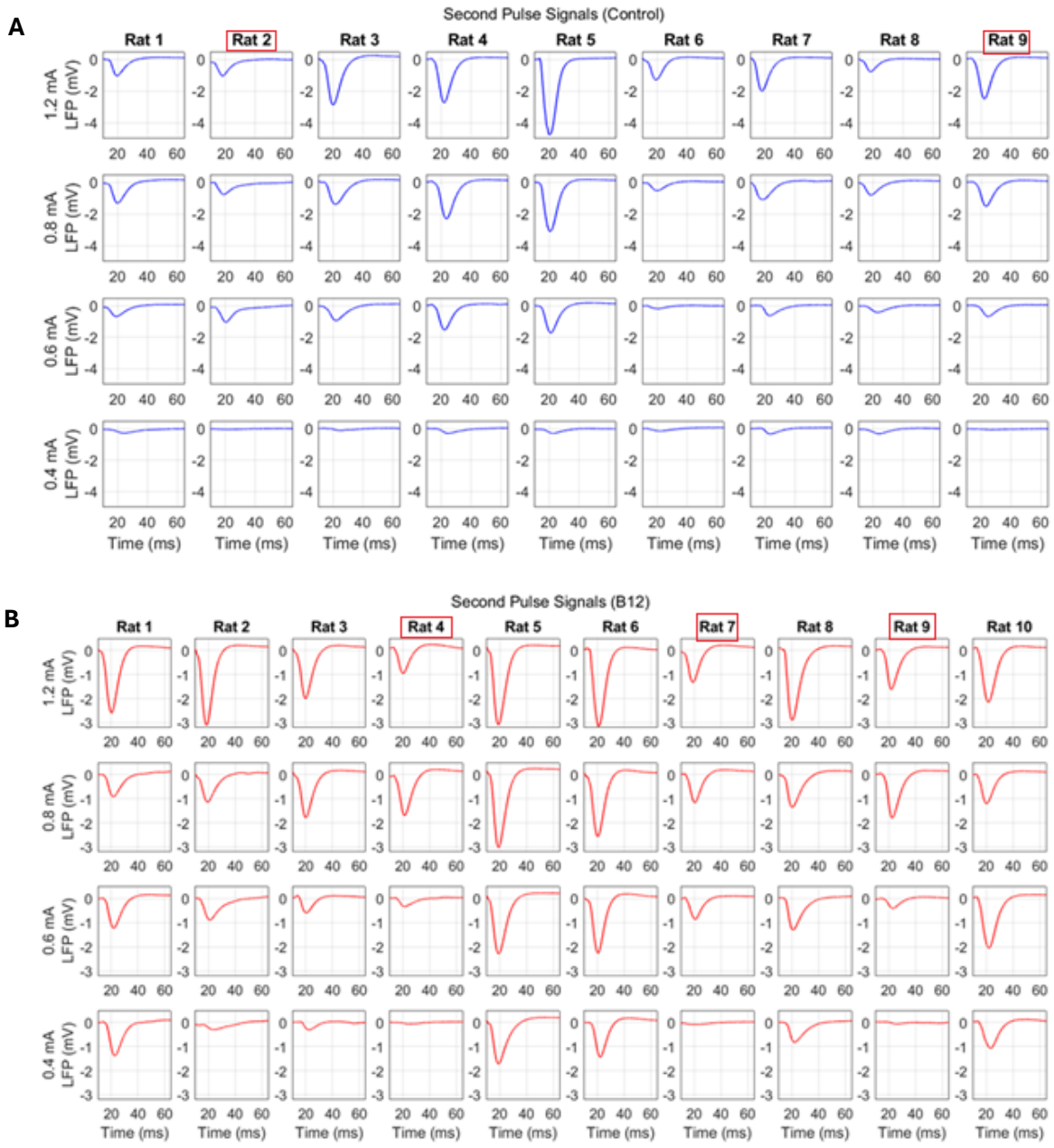


Figure 4. Individual LFP responses during the second pulse phase. *Panel A: control rats (blue); Panel B: B12-supplemented rats (red), across all stimulation intensities. Rats excluded from analysis are indicated by red-boxed titles. Time (ms) is shown on the x-axis, and LFP response (mV) on the y-axis.*

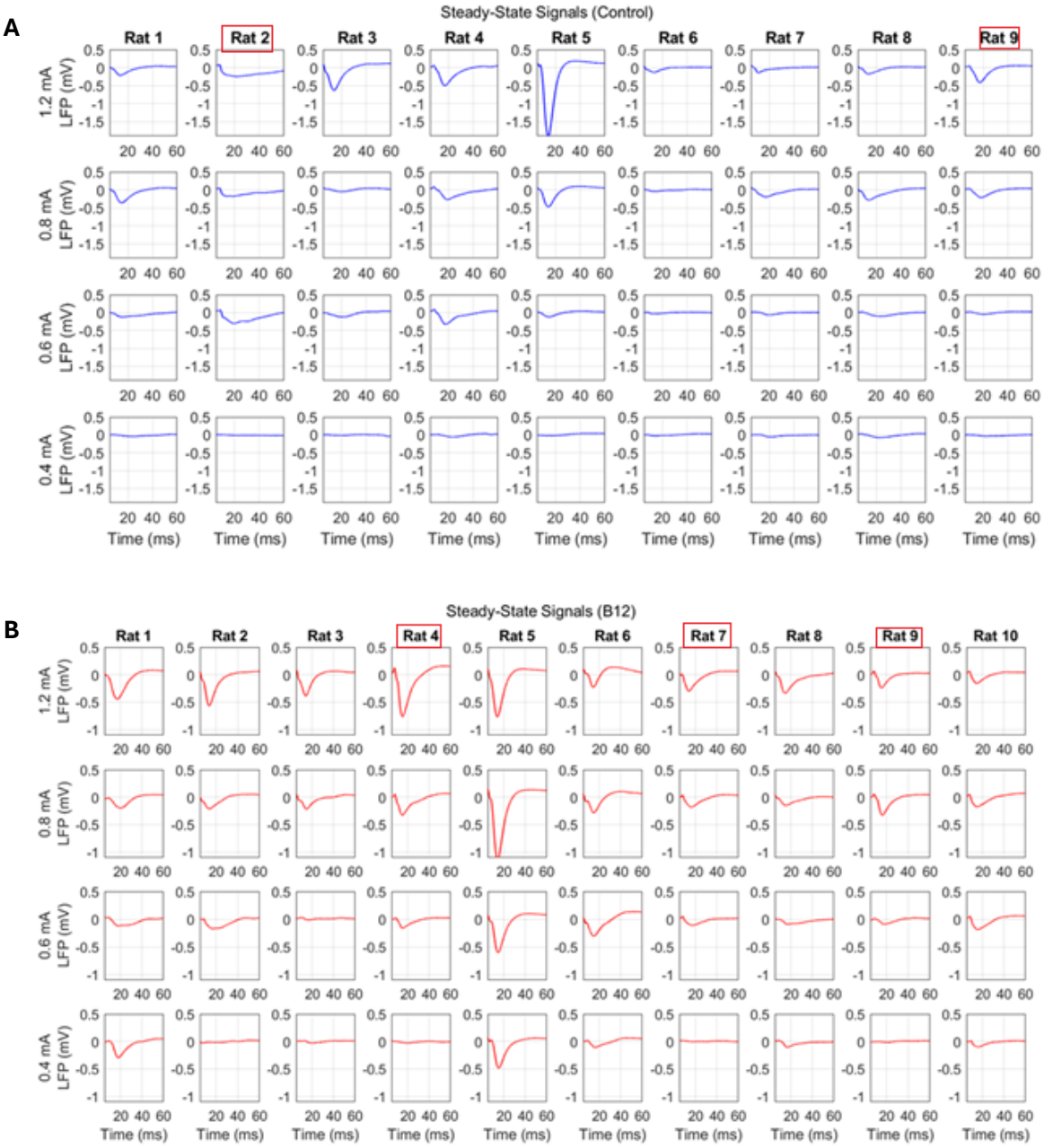


Figure 5. Individual LFP responses during the steady-state phase. *Panel A: control rats (blue); Panel B: B12-supplemented rats (red), across all stimulation intensities. Rats excluded from analysis are indicated by red-boxed titles. Time (ms) is shown on the x-axis, and LFP response (mV) on the y-axis.*

3.4 GROUP-AVERAGED LFP RESPONSES

To assess group-level differences, average LFP responses across all rats are presented for each experimental phase in Figures 6-8. Compared to the control group, the B12 group exhibited larger negative deflections and greater variability, particularly at the lower intensities. As expected, response amplitudes were greatest at 1.2 mA and decreased progressively with lower intensities, while peak latencies remained comparable between groups.

During the first pulse phase (Figure 6), response amplitudes declined with decreasing stimulus intensity in both groups. However, the B12 group showed noticeably larger negative deflections at 0.4-0.8 mA, suggesting enhanced neural responsiveness under lower stimulation. This group also exhibited greater inter-subject variability, as indicated by larger standard errors of the mean (SEMs). Peak latencies were consistent across both groups.

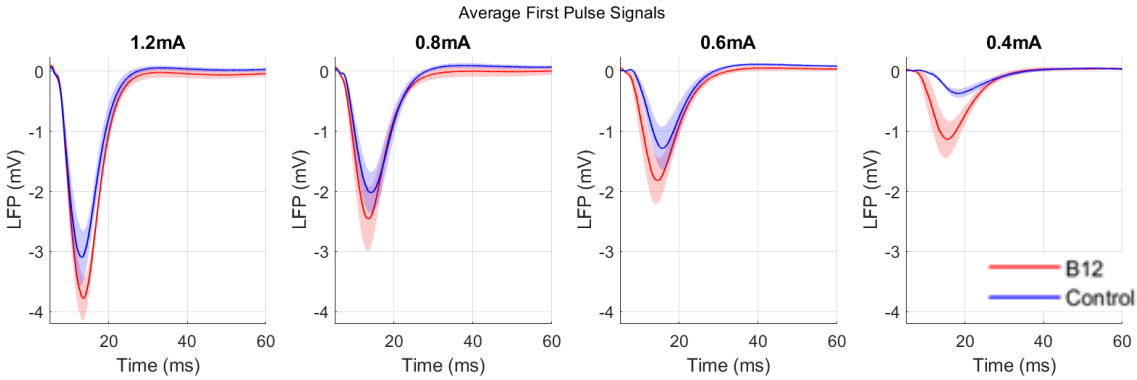


Figure 6. Average first pulse LFP responses across stimulation intensities. *Plots show group-averaged LFP responses for B12-supplemented (red) and control (blue) rats at 1.2, 0.8, 0.6, and 0.4 mA. Shaded regions represent the standard error of the mean (SEM). Time (ms) is shown on the x-axis, and LFP response (mV) on the y-axis.*

Second pulse responses (Figure 7) were attenuated relative to the first pulse, consistent with paired-pulse depression (Sullivan, 2007). The B12 group continued to show stronger negative deflections than controls, particularly at lower intensities, along with increased variability.

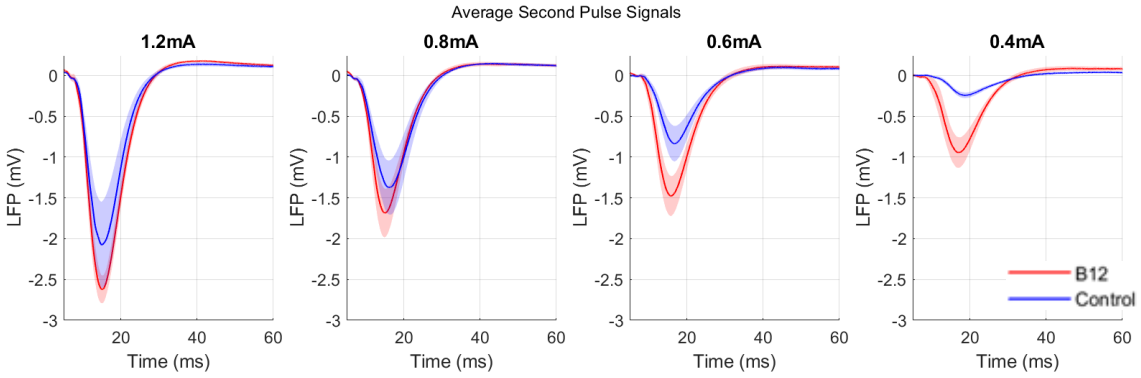


Figure 7. Average second pulse LFP responses across stimulation intensities. *Plots show group-averaged LFP responses for B12-supplemented (red) and control (blue) rats at 1.2, 0.8, 0.6, and 0.4 mA. Shaded regions represent the standard error of the mean (SEM). Time (ms) is shown on the x-axis, and LFP response (mV) on the y-axis.*

Steady-state responses (Figure 8) demonstrated overall amplitude reduction across both groups, indicative of adaptation to repeated stimulation. Notably, the B12 group maintained higher response amplitudes at 1.2 mA and continued to diverge more from the control group at lower intensities, again accompanied by increased variability.

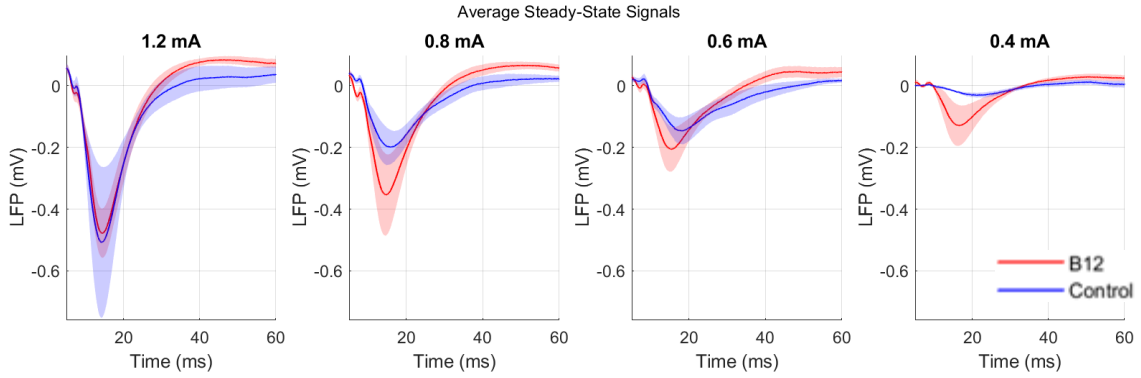


Figure 8. Average steady-state LFP responses across stimulation intensities. *Plots show group-averaged LFP responses for B12-supplemented (red) and control (blue) rats at 1.2, 0.8, 0.6, and 0.4 mA. Shaded regions represent the standard error of the mean (SEM). Time (ms) is shown on the x-axis, and LFP response (mV) on the y-axis.*

3.5 STATISTICAL ANALYSIS OF LFP PARAMETERS

A two-way repeated measures ANOVA was conducted to assess the main effects of group and stimulus intensity, as well as their interaction. Where significant effects were found, post-hoc two-sample t-tests were performed to identify which specific stimulation intensities contributed to the group differences. For these comparisons, effect sizes were calculated using Cohen's *d* to quantify the magnitude of the differences.

3.5.1 Two-Way Repeated Measures ANOVA Results

Table 2. Two-way repeated measures ANOVA results. *P-values are presented for the main effects of group and stimulus intensity, as well as their interaction, across each parameter. P-values are rounded to five significant figures. Statistical significance is denoted by asterisks: p ≤ 0.05 (*), p ≤ 0.01 (**).*

		Peak	P2P	Latency
First Pulse	Group	0.056881	0.047884*	0.12681
	Intensity	1.19E-08**	1.09E-08**	1.41E-07**
	Interaction	0.8608	0.86477	0.27919
Second Pulse	Group	0.012503*	0.009606**	0.19218
	Intensity	9.7E-07**	6.88E-07**	0.22234
	Interaction	0.7913	0.75553	0.48734
Steady-State Response	Group	0.55517	0.65074	0.032161*
	Intensity	0.01018*	0.002134**	0.007497**
	Interaction	0.69016	0.8251	0.013367*

As shown in Table 2, significant group effects were observed for four of the nine parameters: first pulse P2P amplitude (p = 0.0479), second pulse peak (p = 0.0125) and P2P (p = 0.0096) amplitudes - all significantly greater in the B12 group - and steady-state latency (p = 0.0322), which was significantly shorter in the B12 group. Interaction effects were not significant for most parameters, with the exception of steady-state latency. As expected, stimulus intensity had a significant effect on nearly all parameters; however, this was not the primary focus of the study.

3.5.2 Post-Hoc Comparisons and Effect Sizes.

Table 3. Post-hoc two-sample t-test results. *P-values are shown for each stimulation intensity across the four parameters that demonstrated significant group effects in the ANOVA. Significant comparisons are marked by asterisks: p ≤ 0.05 (*), p ≤ 0.01 (**). Cohen's d values are included to indicate effect sizes for significant results. p and d values are values rounded to five significant figures.*

Parameter	Intensity	p-value	d
First Pulse P2P	1.2	0.36875	
	0.8	0.71381	
	0.6	0.35188	
	0.4	0.028053*	1.3349
Second Pulse Peak	1.2	0.38536	
	0.8	0.65265	
	0.6	0.077088	
	0.4	0.003666**	1.9226
Second Pulse P2P	1.2	0.34987	
	0.8	0.65785	
	0.6	0.080217	
	0.4	0.003474*	1.9384
Steady-State Latency	1.2	0.31176	
	0.8	0.63291	
	0.6	0.12681	
	0.4	0.060598	1.1069

To further explore the significant group effects, post-hoc two-sample t-tests were conducted, and Cohen's *d* values were calculated to evaluate effect sizes. As shown in Table 3, statistically significant group differences with large effect sizes ($d > 1.1$) were observed exclusively at the lowest stimulation intensity (0.4 mA). Figures 9-11 illustrate that the B12 group generally exhibited higher amplitudes across all intensities and phases, with the most pronounced differences at 0.4 mA, where non-overlapping error bands indicated statistical significance. In contrast, latency measures showed no consistent group differences (Figures 9C, 10C, and 11C), although steady-state latency approached significance ($p = 0.0606$).

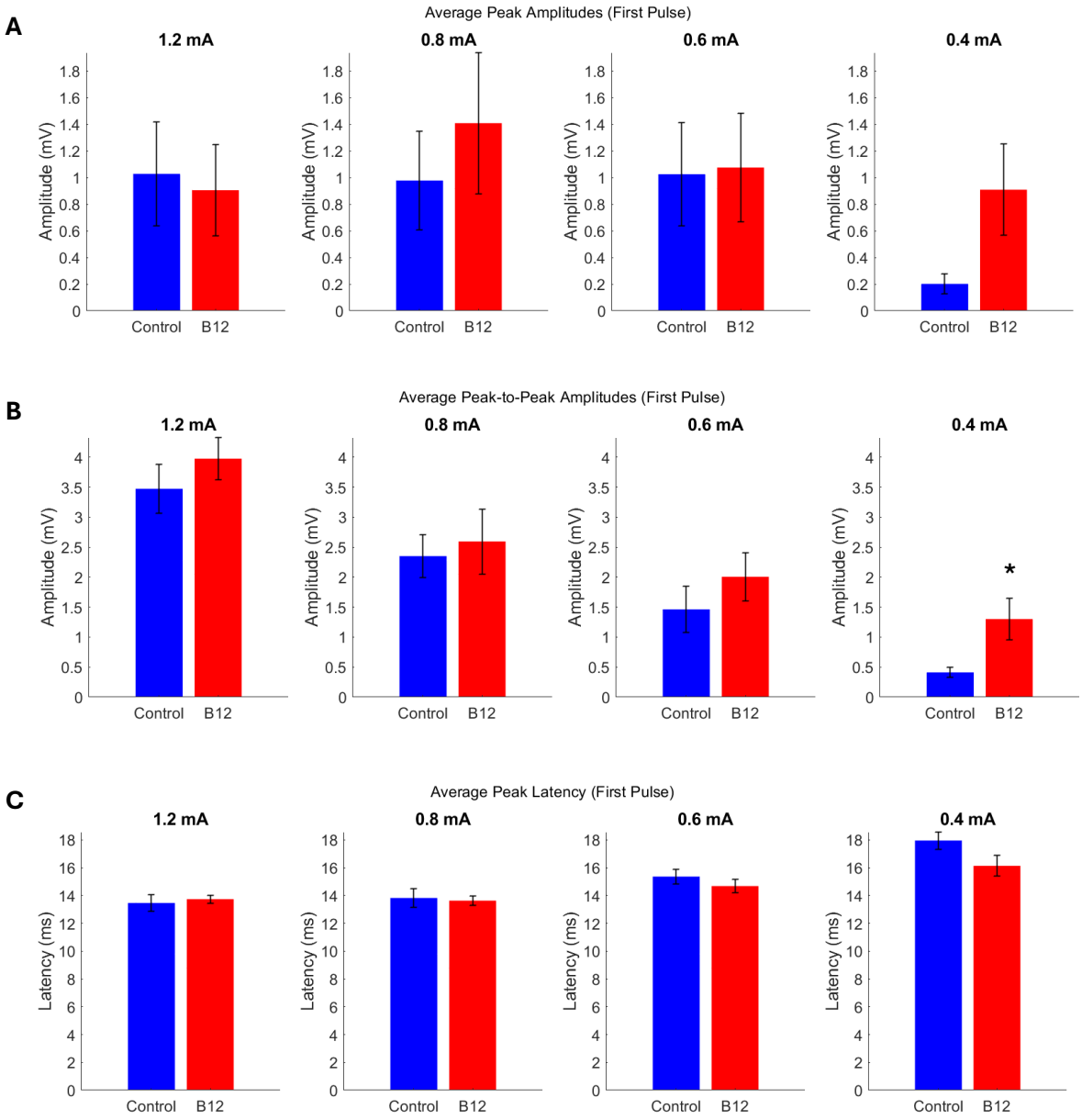


Figure 9. Group comparisons for first pulse parameters. *Bar plots show average values across four stimulation intensities for: (A) peak amplitude, (B) peak-to-peak amplitude, and (C) peak latency. Statistical significance is denoted by $p < 0.05$ (*) and $p < 0.01$ (**). Error bars represent the standard error of the mean (SEM). Control and B12 groups are shown in blue and red, respectively.*

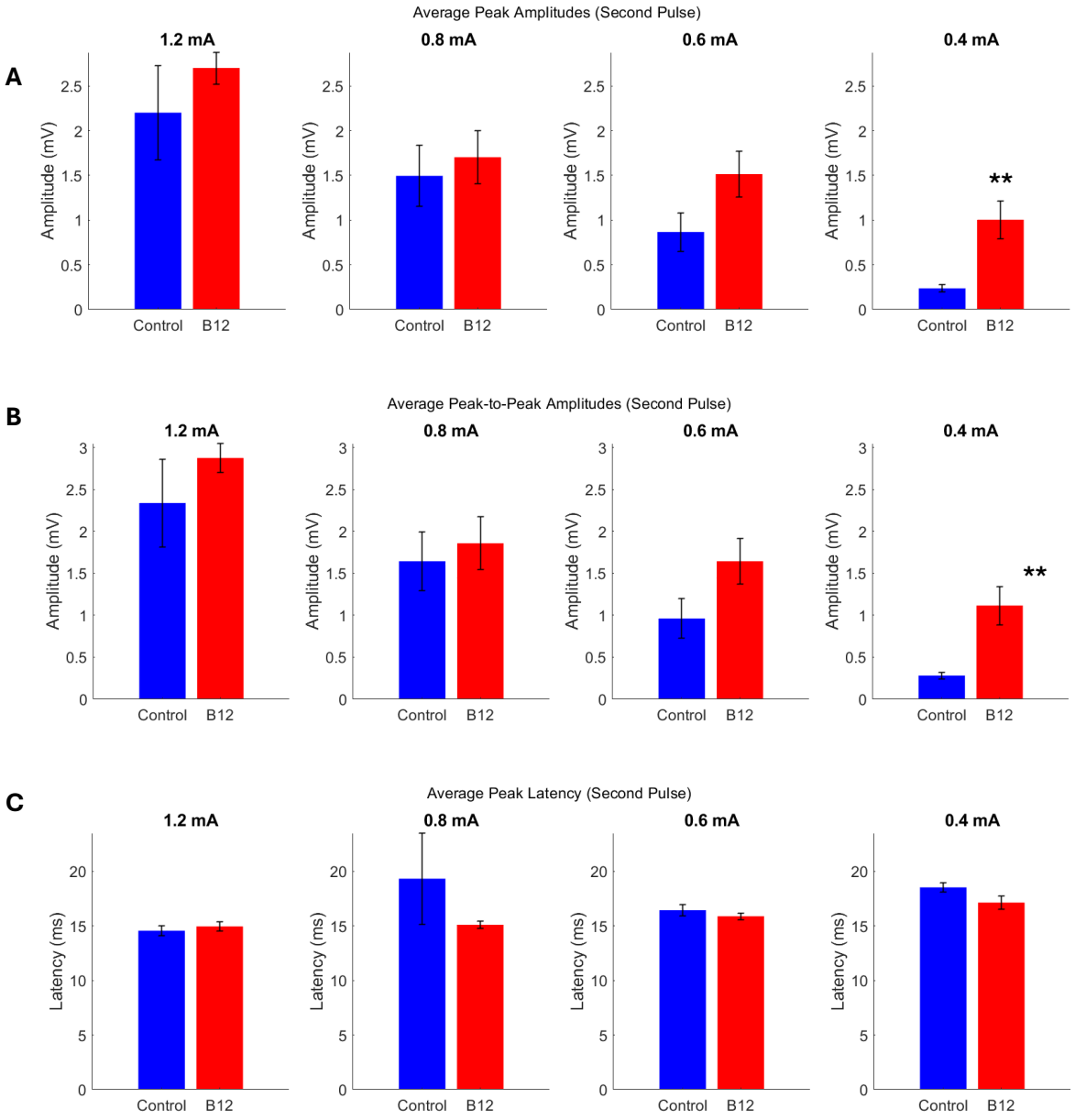


Figure 10. Group comparisons for second pulse parameters. *Bar plots show average values across four stimulation intensities for: (A) peak amplitude, (B) peak-to-peak amplitude, and (C) peak latency. Statistical significance is denoted by $p < 0.05$ (*) and $p < 0.01$ (**). Error bars represent the standard error of the mean (SEM). Control and B12 groups are shown in blue and red, respectively.*

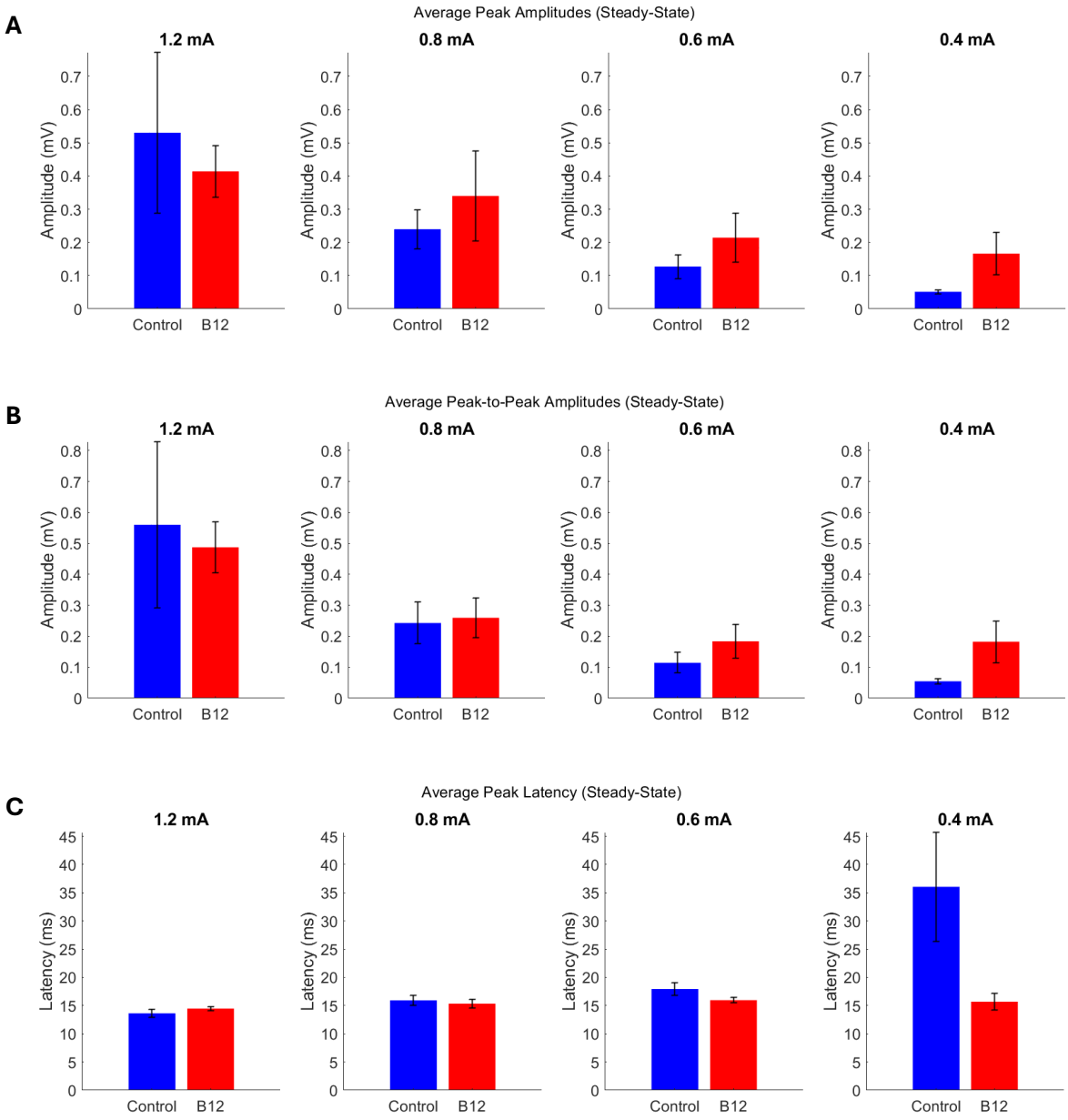


Figure 11. Group comparisons for steady-state parameters. *Bar plots show average values across four stimulation intensities for: (A) peak amplitude, (B) peak-to-peak amplitude, and (C) peak latency. Statistical significance is denoted by $p < 0.05$ (*) and $p < 0.01$ (**). Error bars represent the standard error of the mean (SEM). Control and B12 groups are shown in blue and red, respectively.*

4 DISCUSSION

4.1 SUMMARY OF KEY FINDINGS

This study demonstrates that B12 supplementation significantly modulates local field potentials in the barrel cortex, with the most pronounced effects observed at the lowest stimulus intensity (0.4 mA). Specifically, B12 administration significantly increased first P2P amplitude, as well as second pulse peak and P2P amplitudes, indicating enhanced cortical responsiveness and excitability during both initial and repeated stimulation (Stephani et al., 2021).

In addition to amplitude enhancements, B12 supplementation reduced steady-state latency, suggesting faster or more efficient neural conduction during sustained sensory input (Shelly et al., 2023). The consistently large effect sizes ($d > 1$) highlight the strength and potential biological significance of these findings. Importantly, no significant group differences were found at higher intensities (0.6-1.2 mA), and substantial inter-subject variability was noted in the B12 group. This suggests that B12's neuromodulatory effects are context-dependent and may be influenced by individual differences in neural responsiveness, such that some animals may benefit more from supplementation than others (Cleveland Clinic, 2022).

4.2 COMPARISON WITH EXISTING LITERATURE

4.2.1 B12's Role in Neurophysiology

These findings are consistent with previous research demonstrating B12's contribution to myelination, neurotransmitter synthesis, and synaptic plasticity (Mehrdad, Elyasi & Emsehgol, 2021). The observed reduction in steady-state latency aligns with improved axonal conduction, potentially reflecting enhanced myelin integrity (McDougall et al., 2018). Similarly, modest increases in response amplitudes may reflect enhanced neurotransmitter dynamics, particularly serotonergic modulation of glutamatergic and GABAergic systems (Celada, Puig & Artigas, 2013). This may involve the upregulation of NMDA receptor function or enhanced GABAergic inhibition via increased cofactor availability for neurotransmitter synthesis enzymes (Allen, Sabir & Sharma, 2023).

Although B12 deficiency is known to impair sensory and cognitive processing, the present findings extend this literature by suggesting that B12 supplementation can modulate

neural responsiveness even in animals without deficiency. However, the selective nature of these effects, primarily at lower intensities, draws attention to the subtle, context-dependent nature of B12's impact.

4.2.2 Findings in Context and Unique Contributions

Previous studies have reported general enhancements in neural function following B12 supplementation (Dangour et al., 2015), but the present results indicate that these effects may only emerge under specific stimulus or circuit conditions. The absence of significant group differences at higher stimulation intensities may reflect ceiling effects in cortical excitability or a saturable influence of B12 on circuits that are already maximally engaged.

This context specificity suggests that B12 may act more effectively when circuits are in a subthreshold or partially activated state, rather than uniformly boosting excitability across all conditions. This has important implications for therapeutic application, emphasising the need to account for baseline circuit dynamics when evaluating or designing interventions involving B12 or similar neuromodulators.

4.3 Potential Mechanisms and Explanations

4.3.1 Why Were Effects Strongest at 0.4 mA?

The more consistent effects observed at 0.4 mA may be due to greater sensitivity of low-intensity stimuli to modulation by subtle changes in synaptic efficacy or conduction velocity. In contrast, responses at higher intensities may approach a ceiling, where synaptic and circuit-level activation are already near-maximal (Glazewski & Barth, 2014), thereby limiting the observable impact of B12 supplementation. This saturation could account for the lack of significant group differences at higher stimulation levels.

Alternatively, the observed pattern may indicate a non-linear relationship between stimulus strength and B12 efficacy. Such a relationship warrants further investigation using a broader and more finely graded range of stimulation intensities to better characterise the dose-response profile of B12's neuromodulatory effects.

4.3.2 Why Was the First Pulse Less Affected?

The relatively modest changes observed in first-pulse amplitudes suggest that initial sensory responses may depend more heavily on intrinsic neuronal excitability and fast-conducting pathways, which are less susceptible to modulation by acute B12

supplementation (Alejandre-García et al., 2022). In contrast, the more pronounced effects seen in second-pulse and steady-state responses likely reflect B12's influence on short-term synaptic plasticity mechanisms, such as facilitation and augmentation, where its role in supporting neurotransmitter synthesis and release may play a more prominent role.

Moreover, structural factors such as synaptic density and circuit architecture may inherently limit the extent to which initial responses can be modulated, as these characteristics are relatively stable over short timescales. Conversely, the dynamic synaptic processes engaged during repeated stimulation are more plastic and responsive to acute neuromodulatory influences (Udvary et al., 2022).

4.3.3 Why Did Latency Decrease in Steady-State Responses?

The reduction in steady-state latency observed following B12 supplementation likely reflects two complementary mechanisms. First, enhanced myelination may improve axonal conduction velocity, enabling faster action potential propagation. Second, increased synaptic efficiency, possibly through upregulated neurotransmitter synthesis or heightened receptor sensitivity, could reduce synaptic transmission delays.

These changes would be especially beneficial under sustained stimulation, where rapid conduction speed and precise timing are critical for maintaining phase-locked neural responses (Fedchyshyn & Wang, 2007). The absence of latency changes in single-pulse responses reinforces the idea that B12's effects may be cumulative or activity-dependent, becoming more evident during repetitive or prolonged stimulation.

4.4 Limitations and Future Directions

4.4.1 Study Limitations

While the sample size (n = 7 per group) was sufficient to detect medium-to-large effects, a larger cohort would improve the validity and reproducibility of the results (Andrade, 2020). The exclusion of animals with poor signal quality may introduce selection bias, which could influence the group-level comparisons (Alexander et al., 2015).

Furthermore, the use of anaesthesia during recording is a notable limitation, as anaesthetic agents can suppress cortical excitability and alter synaptic dynamics (Hao et al., 2020), possibly leading to an underestimation of B12's true effects. Despite efforts to check statistical assumptions, the use of parametric tests under some non-normal

conditions could affect interpretation (Vickers, 2005); future analyses incorporating non-parametric methods or mixed-effects models would strengthen the findings.

Finally, this study focused on short-term B12 supplementation. Whether these neural changes persist over time or translate into structural or functional plasticity remains an open question.

4.4.2 Future Research Directions

Future studies should aim to reveal the neurochemical mechanisms by which B12 alters synaptic and conduction properties. Particular attention should be given to neurotransmitter systems such as glutamate, GABA, and acetylcholine, which are key regulators of cortical excitability (Petroff, 2002; Sam & Bordoni, 2023). Molecular profiling and receptor expression analyses could help identify specific targets of B12 action (Piris, 2011).

Longitudinal research is also needed to determine whether the neural enhancements observed with B12 supplementation are sustained over time. Investigating biological variables such as sex, strain, and age would help clarify the generalisability of these effects, especially since the current study was limited to young adult female Lister hooded rats.

Exploring dose-response relationships could establish optimal dosing regimens and help determine whether excessive B12 supplementation has diminishing or adverse effects. In parallel, evaluating the timing of measurements relative to B12 administration would clarify whether its neuromodulatory effects are immediate, delayed, or cumulative.

Finally, future work should employ more advanced methodologies, such as multi-electrode arrays, single-unit recordings, and optogenetics (Simkin et al., 2022; Baars & Gage, 2010; Takeuchi & Berényi, 2020), to dissect how B12 alters neural firing and network dynamics at higher resolution. Integrating these neural measures with behavioural assessments in awake, freely moving animals would bridge the gap between cortical physiology and functional outcomes, greatly enhancing the translational relevance of this research.

4.5 Broader Implications

The findings suggest that B12 supplementation holds therapeutic promise for neurological conditions characterised by sensory deficits or impaired cortical processing.

By enhancing synaptic plasticity and conduction efficiency, B12 could aid individuals with neurodegenerative diseases such as Alzheimer's and Parkinson's, where sensory and cognitive impairments are common.

Moreover, B12 supplementation may benefit individuals with developmental disorders such as autism spectrum disorder and ADHD, both of which involve atypical sensory processing (Marco et al., 2011; Lukovac et al., 2024). Given that populations such as older adults, vegetarians, and individuals with malabsorption disorders are at increased risk for B12 deficiency, targeted supplementation could represent a low-cost, low-risk strategy for promoting cortical function and cognitive resilience.

Taken together, these findings not only advance the understanding of B12's neuromodulatory effects but also highlight its potential as a simple yet powerful tool for supporting brain health across a range of vulnerable populations.

5 CONCLUSION

This study investigated the effects of vitamin B12 supplementation on neural activity within the barrel cortex using whisker pad stimulation and local field potential analysis. The results revealed that B12 supplementation significantly enhanced sensory-evoked cortical responses, particularly at lower stimulus intensities, suggesting improved cortical excitability and neural processing efficiency. These findings reinforce B12's essential role in maintaining optimal neural function and synaptic communication.

Crucially, this work extends the current understanding of B12's neurophysiological impact by highlighting its influence on early sensory processing, a domain less studied compared to its established roles in cognition and neurodegenerative disease. Showing that B12 can modulate sensory-evoked cortical activity opens new avenues for investigating how micronutrient availability shapes neural system function more broadly.

Several limitations must be acknowledged. The modest sample size and use of anaesthesia constrain the generalisability of the findings to broader populations and naturalistic conditions. Future research should assess whether similar effects occur in awake, behaving animals, across different cortical regions, and in varied species. In parallel, mechanistic studies probing how B12 influences synaptic plasticity, neurotransmitter dynamics, and metabolic pathways will be essential for clarifying the biological basis of these effects.

In conclusion, B12 supplementation appears to enhance sensory-evoked cortical processing, potentially improving neural responsiveness within the barrel cortex. These findings lay a foundation for future research into B12's broader role in brain health and support its potential as a therapeutic strategy for alleviating sensory deficits and mitigating neurodegenerative decline.

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